Conditional Elimination and Circular Storage Codes

Hans Brende GPT-6 Astra (OpenAI)

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Abstract

A conditional-elimination inequality gives the weighted storage bound $2h(V) \le \sum_v h(\{v\})$ on every Andrasfai graph and every circular clique of ratio below three. This strictly extends the range at most 14/5 proved in the accompanying fourth paper. The new family bound, homomorphism transfer, and common-alphabet cardinality corollary are verified in Lean, using that paper's existing definitions. An exact conditional-mutual-information identity and the unequal-alphabet corollary have complete written proofs. The result does not supply these graph hypotheses for general native tape cones.

Contributions. Hans Brende directed the research. GPT-6 Astra (OpenAI) developed the new argument, formalization, and text under his direction. The memo reuses the fourth paper's framework and results, by Hans Brende and Claude Opus 5.5 (Anthropic).

1 The strengthening and its scope

The fourth paper proves that every storage-valid polymatroid on a circular clique $K_{p/q}$ with $p/q \le 14/5$ satisfies a weighted half bound [2, Proposition "Further good targets"]. This memo extends that range to $p/q < 3$. The exact endpoint fails on K_3. The existing paper and its Lean declarations remain unchanged; the memo supplies a general proof and derives the earlier range from it.

A polymatroid here is a function $h : 2^V \to \mathbb{R}$ on a finite set, with $h(\varnothing) = 0$, monotonicity, and submodularity. Write

$$h(A \mid S) = h(A \cup S) - h(S), \qquad I_h(A; B \mid S) = h(A \mid S) + h(B \mid S) - h(A \cup B \mid S).$$

The mutual information is nonnegative for the disjoint sets used below. Validity for recovery sets $N(v)$ means $h(v \mid N(v)) = 0$. Diminishing returns preserves this equality after enlarging the conditioning set. For a graph, $N(v)$ is its neighborhood. No representability, integrality, or equality of singleton values is assumed.

2 Conditional elimination

Theorem 2.1 (An exact partition identity)**.** Not formalized; a complete proof is provided.

Let $P_1, \dots, P_m$ partition V, where $m \ge 2$ and empty parts are allowed. Enumerate V as $v_0, \dots, v_{n-1}$ and put $S_t = \{v_0, \dots, v_{t-1}\}$, $P_i^t = P_i \setminus S_t$. Let $a(t)$ be the part of v_t. Suppose there is $b(t) \ne a(t)$ with

$$h(v_t \mid S_t \cup P_{b(t)}^t) = 0.$$

Then

$$\sum_i h(P_i) - 2h(V) = \sum_{t=0}^{n-1} \sum_{i \notin \{a(t), b(t)\}} I_h(v_t; P_i^t \mid S_t) \geq 0. \tag{2.1}$$

In particular $2h(V) \leq \sum_i h(P_i) \leq \sum_{v \in V} h(\{v\})$.

Proof (not formalized). Set $J_t = \sum_i h(P_i^t \mid S_t) - 2h(V \setminus S_t \mid S_t)$. Fix a step and abbreviate its vertex, own part, supplying part, and conditional entropy by $v, a, b, u = h(v \mid S_t)$. The own-part contribution decreases by u. Each other part decreases by $I_h(v; P_i^t \mid S_t)$, and the whole remaining entropy decreases by u. Thus

$$J_t - J_{t+1} = \sum_{i \neq a} I_h(v; P_i^t \mid S_t) - u.$$

The supplying-part condition makes its summand exactly u. Cancel it and telescope, using $J_n = 0$ and $J_0 = \sum_i h(P_i) - 2h(V)$. Nonnegative mutual information and subadditivity within the parts give the two inequalities. The empty ground set and zero conditional entropies cause no exception. □

More explicitly, the weighted slack is the right side of (2.1) plus $\sum_i (\sum_{v \in P_i} h(\{v\}) - h(P_i))$, another sum of nonnegative mutual informations. If the supplying condition is dropped, the same calculation subtracts $\sum_t h(v_t \mid S_t \cup P_{b(t)}^t)$ from the right side. That is an exact defect identity, not an assertion that a low-defect order always exists.

The following equivalent style of elimination is the short formal interface used for the graph application. It is stated separately because Lean verifies its inequality, not the preceding general multi-part identity.

Lemma 2.2 (A three-record criterion)**.** `Lean: three_part_elimination` [`Elimination.lean`]
Let $A, B, C \subseteq V$, not necessarily disjoint. Suppose that for every proper revealed set $S \subsetneq V$ there is $v \notin S$ for which one of the following holds:

$$\begin{aligned} v \in A, &\quad h(v \mid S \cup B) = 0; \\ v \in B, &\quad h(v \mid S \cup C) = 0; \\ v \in C, &\quad h(v \mid S \cup A) = 0. \end{aligned}$$

Then $2h(V) \leq h(A) + h(B) + h(C)$.

Outline of the formal proof. Induct downwards on the number of unrevealed vertices to prove $2h(V) + h(S) \leq h(S \cup A) + h(S \cup B) + h(S \cup C)$. Adding an eligible v changes neither the own-record nor the supplying-record entropy. Submodularity bounds the increase of the third record by $h(S \cup \{v\}) - h(S)$. The induction therefore passes from $S \cup \{v\}$ to S. At $S = V$ it is equality; at $S = \varnothing$ it is the conclusion. □

For a proper partition into three independent graph parts A, B, C, orient edges cyclically $A \to B \to C \to A$. If this orientation is acyclic, it satisfies the criterion: in each unrevealed induced graph choose a source. Its unrevealed neighbors all belong to the next part; its other neighbors are already in S. Validity and diminishing returns give the displayed dependence. This passage from an arbitrary acyclic orientation is a written argument. The specific order for the Andrasfai application below is checked directly in Lean.

Acyclicity cannot be omitted from the three-record claim. On a six-cycle colored A, B, C, A, B, C, the binary word $(U, V, U+V, U, V, U+V)$ for independent fair bits is recoverable at every vertex. Its joint entropy is two, but each of the three color records has entropy one. This does not violate the weighted half bound for the six-cycle; it distinguishes that bound from this particular three-record inequality.

3 All circular ratios below three

The circular clique $K_{p/q}$ has vertices $0, \dots, p-1$, with adjacency given by $q \le (j-i) \bmod p \le p-q$. For $k \ge 2$ put $\mathrm{And}(k) = K_{(3k-1)/k}$.

Theorem 3.1 (All Andrasfai graphs)**.** Lean: `rateBound_andrasfai` [`Circular.lean`]

Every valid polymatroid on $\mathrm{And}(k)$*,* $k \ge 2$*, satisfies* $2h(V) \le \sum_{v \in V} h(\{v\})$*.*

Outline of the formal proof. Use independent intervals $A_i = i$, $B_j = k + j$ for $0 \le i, j < k$, and $C_\ell = 2k + \ell$ for $0 \le \ell < k - 1$. Directly from circular adjacency,

$$A_i \sim B_j \iff i \le j, \quad B_j \sim C_\ell \iff j \le \ell, \quad C_\ell \sim A_i \iff \ell < i.$$

The order $A_0, B_0, C_0, A_1, B_1, C_1, \dots, A_{k-1}, B_{k-1}$ places every cyclically oriented edge forward. Lean checks these inequalities using the integer order keys $3i, 3j + 1, 3\ell + 2$. For any unrevealed set take its least key; all later neighbors lie in the next part. Lemma 2.2 gives $2h(V) \le h(A) + h(B) + h(C)$, and subadditivity over the partition gives the weighted assertion. □

Corollary 3.2 (Circular targets and homomorphisms)**.** Lean: `rateBound_circularClique_lt_three`, `rateBound_of_circularHom`, `rateBound_circularClique_oldRange` [`Circular.lean`]

For integers $q > 0$ *and* $0 \le p < 3q$*, every valid polymatroid on* $K_{p/q}$ *satisfies the weighted half bound. The same holds on every finite simple graph with a homomorphism to such a clique. The previous range* $5p \le 14q$*, including its degenerate* $p = q = 0$ *case under the Lean empty-graph convention, is a corollary.*

Outline of the formal proof. Choose $k = q + 1 \ge 2$. Since $p \le 3q - 1$, $pk \le (3k - 1)q$. The already formalized Bondy–Hell map $i \mapsto \lfloor (3k-1)i/p \rfloor$ gives $K_{p/q} \to \mathrm{And}(k)$ [1]. For $p = 0$ this is the unique empty-domain map; no division is evaluated at a vertex. Apply Theorem 3.1 and the fourth paper's homomorphism pullback. That pullback sends h to $Z \mapsto h(\phi^{-1}(Z))$ and uses subadditivity in each fibre. If $q > 0$ and $5p \le 14q$, then $p < 3q$; if $q = 0$, the older range forces $p = 0$ and the conclusion is the empty-graph bound. No earlier numerical circular-clique certificate is used in this proof. □

Corollary 3.3 (Arbitrary common alphabets)**.** Lean: `card_sq_le_of_circularHom` [`Codes.lean`]

Let G *be a finite simple graph mapping to* $K_{p/q}$ *with* $q > 0$ *and* $p < 3q$*. For any nonempty finite alphabet* $\mathcal{A}$*, every storage code* $\mathcal{C} \subseteq \mathcal{A}^{V(G)}$ *satisfies* $|\mathcal{C}|^2 \le |\mathcal{A}|^{|V(G)|}$*. No linearity assumption is imposed.*

Outline of the formal proof. Supply Corollary 3.2 to the fourth paper's existing cardinality theorem. That theorem applies the weighted bound to the entropy of a uniform codeword and converts the logarithmic inequality to the stated integer one. It handles empty codes and one-symbol alphabets as well. No entropy library is duplicated. □

Corollary 3.4 (Unequal alphabets)**.** Not formalized; a complete proof is provided.

On the same graphs, every storage code $\mathcal{C} \subseteq \prod_v \mathcal{A}_v$ *over nonempty finite alphabets satisfies* $|\mathcal{C}|^2 \le \prod_v |\mathcal{A}_v|$*.*

Proof (not formalized). For a nonempty code, take its uniform law. Its entropy is a valid polymatroid and has $h(V) = \log |\mathcal{C}|$ and $h(\{v\}) \le \log |\mathcal{A}_v|$. Apply Corollary 3.2 and exponentiate. Empty codes satisfy the inequality directly. □

The endpoint is sharp: $K_{3/1} = K_3$ supports the binary parity code $(U, V, U + V)$ of size four, whose square exceeds 2^3. The half constant is also attained on each Andrasfai graph with a suitable alphabet: give each edge an independent a-ary symbol and store its incident tuple at each vertex. The graph is k-regular, so the coordinate alphabet has size a^k, and $2|E| = k|V|$ gives rate exactly one half. This is the standard edge-symbol construction, not a new encoding method.

Contribution to the program. The memo widens an all-alphabet impossibility tool and replaces isolated circular-target certificates by one explicit elimination mechanism. A prospective construction on these graphs cannot exceed half rate. An application to native transposition still needs a proved graph homomorphism or elimination interface for its actual recovery relation; neither is furnished here. The general tape question remains open. The argument's correctness and formal verification are separate from external priority, which has not been established for this strengthening.

A Index of Lean declarations

Declarations are relative to `CircularMemo`; their axiom checks are generated together with this index (Appendix B).

`CircularMemo.Circular` `rateBound_andrasfai`, `rateBound_circularClique_lt_three`, `rateBound_circularClique_oldRange`, `rateBound_of_circularHom`.

`CircularMemo.Codes` `card_sq_le_of_circularHom`.

`CircularMemo.Elimination` `three_part_elimination`.

B Verification and reuse

The authoritative proofs of the results marked `Lean` are in the memo's `lean/CircularMemo/` library. The English outlines describe those proofs for convenience. The general partition identity, the general acyclic-orientation argument, unequal alphabets, and the sharpness examples have written proofs; they are not claimed to be formalized by the three-part inequality alone.

The library requires the fourth paper by the relative path `../../lean`; that paper in turn requires the earlier shared libraries. Clone the whole repository, preserving sibling directories. Lean and Mathlib remain pinned to v4.34.1. No new axiom, proof placeholder, or kernel-bypassing decision procedure is used. The permitted axiom set is `propext`, `Classical.choice`, and `Quot.sound`.

From the memo directory, run `python3 tools/lean_index.py`. It generates this declaration index and `lean/PaperAxioms.lean` from the same formal-status annotations. In `lean/`, run `lake build` and `lake env lean PaperAxioms.lean`. On a fresh clone first fetch the pinned dependency cache with `lake exe cache get`. In `paper/`, run `latexmk -pdf main.tex`.

The main paper's original declarations and certificate remain intact. The memo's old-range corollary derives that range directly from the stronger theorem, including the empty case. Removing the old certificate from its original module is deferred until a separate dependency and publication review.

References

[1] J. A. Bondy and Pavol Hell. A note on the star chromatic number. *Journal of Graph Theory*, 14(4):479–482, 1990. doi:10.1002/jgt.3190140412.

[2] Hans Brende and Claude Opus 5.5 (Anthropic). Curved motion, depth reduction, and storage codes for matrix transposition on tapes. Manuscript and partial Lean formalization, repository HansBrende/papers, 2026. Pinned revision a0f97448b0df08822c25948c2cbbdc7dc5a4ed54. URL: https://github.com/HansBrende/papers/tree/a0f97448b0df08822c25948c2cbbdc7dc5a4ed54/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes.